
The Chicago Crimiviz

Process Book

Julien Erbland • Mathis Richard • Max Henrotin

EPFL, Data Visualization, spring 2026

1 From idea to scope

Chicago already has crime tools, and most of them work well. But they are built for analysts: dense, utilitarian, and unwelcoming to anyone who isn't one. We wanted to make something else: a site for the curious, where the same rich data is approachable, easy to explore, and genuinely beautiful to look at.

A useful way to think about the structure of the website is a guided tour of a monument. The home page is the guide that points you to the right room. The Map tab is for the visitors who want to walk around and look. The Trends tab is for the small facts and figures on the way. The Insights tab is the history panel that explains what happened, when, and why it mattered.

The dataset made this possible. The Chicago Police Department publishes every reported incident since 2001 through the city's Socrata portal, also mirrored on Kaggle. The file holds about 8.5 million rows. Each row has a primary crime type, a location description, coordinates, and a flag for whether an arrest was made. The data is clean, the categories are stable, the coverage is long, and the file updates daily, which kept the option of a live feed open for later. With one source we had enough to tell both a spatial story and a temporal one without adding a second dataset.

We checked similar open datasets for New York (311 complaints) and San Francisco (police calls). Neither matched the Chicago file on quality or size: the categories were less consistent, the coordinate precision was lower and less consistent, and the time range was shorter. Chicago was the best choice available.

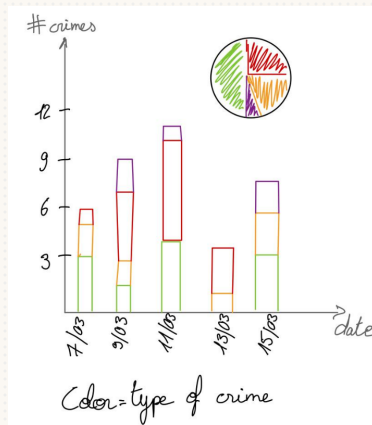
2 Data and feasibility

The first problem was the size of the dataset. A browser cannot draw 8.5 million points at once, so a simple marker map was not an option. We had two practical choices: group the points into administrative areas, or use hex bins. We chose Chicago's **77 community areas**. They are small enough to show real differences between neighbourhoods, but large enough that each one has a clear colour without too much noise.

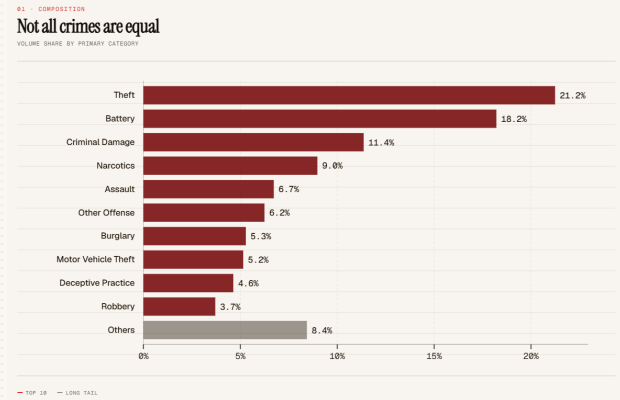
The rest of the cleanup was simple. About 1.1% of rows have no coordinates, so we removed them from the map. We kept them for the time-based charts, where the location does not matter. The map uses about 7.8 million incidents, while the temporal charts use the full 8.5 million. Dates came in the format MM/DD/YYYY HH:MM:SS AM/PM and had to be converted to standard timestamps. The only real design choice was how to handle the crime categories. The dataset has 34 primary categories (burglary, criminal damage, weapons...), which a person can read through, and 569 detailed descriptions, which is too many. We filter and colour the map on the primary category and keep the detailed description in the tooltip.

3 Sketches from Milestone 1, expanded

The Milestone 1 sketches had a hexbin map and a radial "threat clock" next to it. Both changed. The threat clock moved to the optional pile because it did not carry enough on its own to earn a panel and charts were more readable (see Figure 1). The hexbin became a community-area choropleth, for two reasons. First, hexbins gave us a regular grid that looked clean but did not tell anyone where they were looking. With recognisable boundaries, a viewer who does not already know Chicago's geography can anchor the data to the city geography. Community areas, the official seventy-seven neighbourhoods Chicagoans actually refer to, do that anchoring work. Second, the community-area split turned out to be useful well beyond the map itself: by aggregating crime counts per area, we could pre-compute data slices that the map could load efficiently rather than reading from one massive flat file. The choice that made the map readable also made it performant. We also moved from one busy page to four tabs: Home, Map, Trends, Insights. This way, each tab has more space and possibilities to explore their parts as shown in Figure 2.



(a) Trends page sketch



(b) Example of one final trend graph

Figure 1: Implementation of the Trend Tab

4 Design decisions

4.1 Editorial direction over generic dashboard

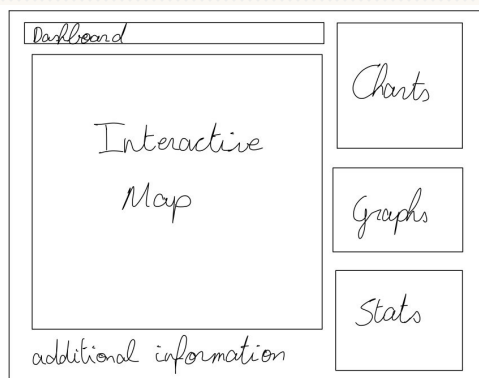
Our first build was a fairly conventional site: a clean dashboard that did the job but looked like every other crime dashboard online. Instead of polishing something forgettable, we redrew the entire visual direction from scratch around one deliberate idea: treat the site as a newspaper about the city's incidents. The masthead reads "The Chicago Crimiviz," each tab is a section of the paper, and the headline counts are set in a newsroom typeface. This was a conscious stance. We wanted the site to carry a point of view rather than the neutral, interchangeable tone of a dashboard.

The boldest part of that direction is a signature of our own: blood that drips down the home page as the user scrolls, and a splatter across the page. It is an unusual and frankly risky choice for a data project, but the subject is violence, and a perfectly sterile design would quietly understate that. It is also what makes the site feel original rather than generic.

The site is built to feel alive rather than static. Navigation cards respond to the cursor, their imagery zooming gently on hover; the insight cards lift off the page and their arrows slide forward; charts, filters and buttons all react as the pointer moves across them. Together with the scroll-driven drips, these small motions give the whole site a responsive, dynamic feel that a flat dashboard never has. All of it respects the **prefers-reduced-motion** accessibility setting: when a visitor has asked their operating system to reduce motion, the site detects it and switches every animation off, drips included.

4.2 Palette in oklch

Every colour in the site is defined using `oklch()` rather than hex codes, for two reasons. First, `oklch` lightness is perceptually even, which means the four greys in the `-ink` scale look equally spaced to the eye even though they all sit on a warm hue. Second, the accent colour (a deep newsprint red) reaches AA contrast against the off-white background without manual adjustment. The result is a palette that stays consistent and accessible across the whole site, defined once and reused everywhere.



(a) Initial Sketch of our website display



(b) Final Home Page with links to the three tabs

Figure 2: Website organization

4.3 Pre-aggregated JSON over a live API

Our first attempt fetched the data live from the Socrata API on every page load. It was a poor experience: each interaction waited on a network round-trip, the site felt sluggish and relied on the live API. We dropped it.

Instead, we took a single snapshot of the dataset on 30 April 2026 and pre-processed everything at build time. A Python script issues a handful of `$group` queries and writes small aggregate files for the choropleth and data analysis. The full incident history is then split out of the source CSV into one GeoJSON per community area: 78 files for the 77 areas, because Austin (CA 25) is too large for GitHub’s file limit and had to be split in two. Every file is hosted directly on GitHub, with no backend and no external dependency at runtime.

The payoff is speed. The front end fetches only the file for the area the user clicks, so dropping into a neighbourhood’s block-level points is near-instant. On data-fetch speed when zooming into individual incidents, our site is among the fastest compared to similar tools, and it carries no rate limit to hit during normal use. We can refresh the whole snapshot by running the script again when desired.

5 Technical implementation

The site is a single static page with no bundler and no framework, backed by a small set of native ES modules under `assets/js/`. The map is built with MapLibre GL JS v4, while the Trends and Insights charts use D3 v7 with topojson-client; both libraries load from a CDN through a native `<script type="importmap">`. The `data/` folder holds the pre-aggregated JSON and the per-area GeoJSON, so the site reads everything statically with no build step.

The CSS is split across three files by responsibility rather than by feature. `main.css` holds the design tokens, reset, layout and media queries, the shell of the page; `components.css` holds each section’s components; and `viz.css` holds the editorial animations and the blood-drip system. The split lets us rework the design without touching the layout shell.

One detail worth recording: the blood filters and gradients live in an inline `<svg class="blood-defs">` at the top of the body rather than in a separate file, because the CSS references them through `filter: url(#bloodRough)` and those URLs are document-local. Moving them out would silently strip every editorial motif.

The JavaScript keeps a single shared `state` object (crime type, hour and years) in `filters.js`, which also owns the input bindings and a small change dispatcher. Every visualisation module imports that state as read-only and redraws when it changes. A short `loadJSON` helper in `data.js` fetches and caches each data file once, and `main.js` boots the page, wiring the tabs, the filters and the scroll-driven animations

6 Map

LED BY MAX HENROTIN

The Map is the interactive part of the site. It opens on the whole of Chicago as a single filterable choropleth: each of the 77 community areas is shaded by how much crime it carries under the filters you set. From there the map works like a focus lens. Click any neighbourhood and it zooms in, drops the choropleth, and loads that area’s complete, unsampled history as a living layer of hotspots and block-level points showcasing each record with details when zooming further (see Figure 3). A “Back to Chicago” button brings you back to the city and a live scale indicator in the bottom-right corner tracks the zoom, from about 1:107,000 at city scale down to roughly 1:105 at street level.

What makes it feel alive is the filtering. Three controls in the sidebar, crime type, hour of day and year, redraw the map as you change them. The year and hour sliders each carry a play button: press it and the two handles collapse into one and sweep across the range, animating the city year by year or hour by hour while the colours shift underneath. Only one plays at a time, so the story stays readable.

6.1 Choropleth over hexbin

We chose a community-area choropleth over a hexbin map. Equal-area hexbins would represent raw spatial density more faithfully, since community areas vary a lot in size, but they look very messy over the whole city. The community-area boundaries give the viewer a clear label for each region (Loop, Lincoln Park, Englewood, and so on), which makes the map easier to read. Nonetheless, when clicking on a community area, we do bin space into a fine grid for the density layer. We simply kept that binning out of the city overview, where orientation matters more than raw uniformity.

6.2 Leaflet, then D3, then MapLibre, then focus mode

The map went through four versions before we settled on the final one.

The first version used Leaflet as a simple base. We replaced it with a pure D3 and TopoJSON choropleth on the 77 community areas, which was clean and lightweight but only worked at one level of detail. The third version used MapLibre with semantic zoom: a choropleth that faded into a heatmap and then into individual points as the user zoomed in. This worked in principle but was hard to read in practice. At street level the city-wide point layer was a wall of overlapping dots, and the 250,000-point sample behind it silently dropped data we wanted to keep.

The final version is **focus mode**. At the city zoom level the user sees a choropleth coloured by volume, with no points and a light hover outline. Clicking an area zooms the map to that area’s boundary and loads the full history for that neighbourhood on demand. The choropleth fades out, and a density layer and block-level points take over. A “Back to Chicago” button returns the user to the full city view. This design gave us three benefits.

All incidents are kept on disk. Each community area has its own GeoJSON file at `data/points/{ca}.geojson`, and the front end only loads the file for the area the user clicks. The full 7.8 million mapped incidents are stored, with no sampling. The only area large enough to exceed GitHub’s file limit is Austin, with about 478,000 incidents, which we split by year. A manifest hides the split from the front end.

The heatmap uses a quantile colour scale. A standard heatmap saturated to dark red almost everywhere in Chicago, because density is high across most of the city. We replaced it with a 50m grid. We count the incidents in each cell, rank the non-empty cells by percentile, and draw the centres as blurred circles using an interpolated colour ramp. The lowest 10% of cells appear near white, and the top 10% appear dark red, regardless of the area’s total volume.

Block-level aggregation. CPD rounds every location to the nearest block centroid for privacy, so many incidents share the same coordinates. In one area, 50,000 incidents share about 7,100 unique points, and a single corner can hold around 950 of them. Rather than draw 950 overlapping dots, we draw one circle per block, sized by the count. At high zoom the count is printed inside the circle, and the popup labels the location as a block centroid rather than a street address.



Figure 3: General View of the Map

6.3 Filters

The map carries three filters, primary crime type, hour of day and year, and all three redraw the choropleth. We trimmed the year range to 2002–2026 because only about 1.2% of 2001 records carry usable coordinates (CPD did not geocode systematically before 2003). Ward, district and arrest filters would multiply combinations without adding much to the spatial view, so we left them out: we opted for a user-friendly interface rather than a technical tool. There is an intuitive filter summary and a reset option displayed on the map.

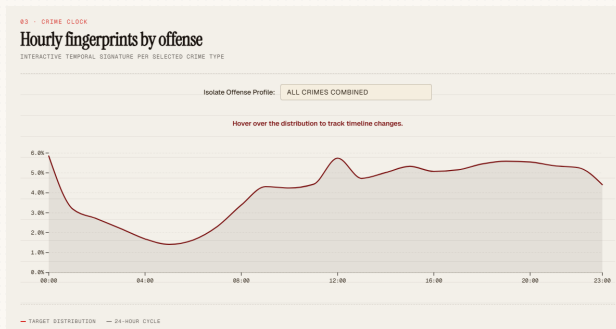
7 Trends

LED BY MATHIS RICHARD

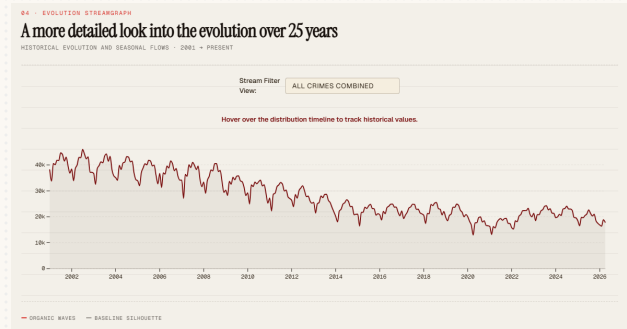
The first component, designated as the Interactive Crime Clock, charts the chronological distribution of incidents across a standard twenty-four hour cycle. By showcasing it horizontally, we can allow a bigger chart filling the page more ‘naturally’. It computes the percentage share of daily volume attributable to each hour, establishing a clear baseline for citywide activity peaks. An integrated dropdown interface allows users to isolate specific offense types, instantly recalculating the corresponding curve. This architecture exposes how different categories possess distinct temporal signatures, contrasting opportunistic street crimes with nocturnal

infractions.

The second module is the Historical Evolution Trend-line, which shifts the analytical window to a longitudinal scale spanning from 2001 to 2026. This component maps absolute monthly incident counts over a continuous quarter-century timeline, highlighting cyclical peaks and valleys. To handle potential structural variations across the source files, the ingestion engine standardizes string values and applies mathematical weighting rules derived from primary offense shares. The resulting widescreen line graph features active crosshair tracking, letting users isolate macro volumes or individual categories to inspect how specific legal, economic, or environmental factors shaped real-world trends over decades.



(a) Time-of-day distribution

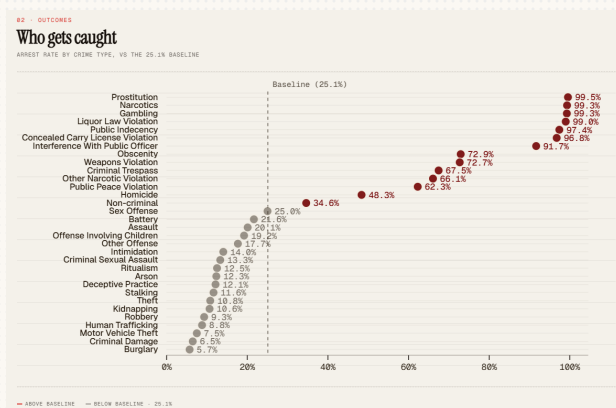


(b) Seasonality over the years

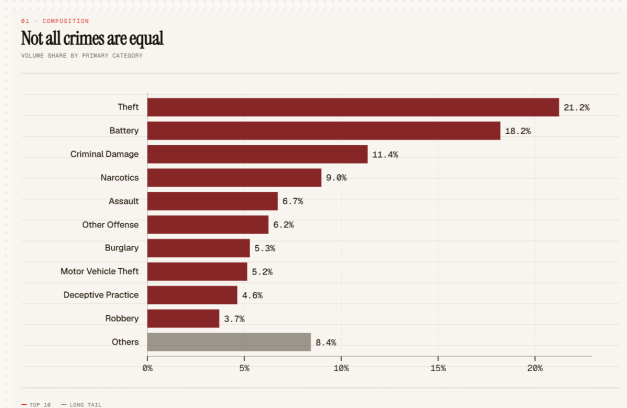
Figure 4: Overall Crime Timeline Analysis

The third component is the Enforcement Outcome Graph, which shifts the analytical window to a comparative evaluation of legal accountability. This component maps categorical arrest rates against a stable 25.1 percent citywide baseline, highlighting systemic variances across different offense types. To handle potential reporting variations across the source files, the ingestion engine groups primary classifications and filters out procedural noise. The resulting widescreen dot plot features active crosshair tracking, letting users isolate specific crime categories to inspect how proactive field operations or reactive investigations shape real-world enforcement trends.

Finally, the last graph evaluates the structural layout through a Composition Visualization. This module addresses the categorical distribution of the 8.5 million rows by partitioning incidents into primary offense classifications. It isolates high-volume crime categories from the extensive long-tail distribution, formatting the findings into an editorial bar chart that matches the vintage newspaper aesthetic of the overall site. Together, these three lenses convert a dense, multi-dimensional database into an accessible layout, balancing proactive historical tracking with reactive daily insights.



(a) Arrest Rate depending on crime



(b) Different Crime Volumes

Figure 5: Statistics regarding numbers and arrests

8 Insights

LED BY JULIEN ERBLAND

The Insights tab explores how the crime data reacted to specific events and crises that happened in Chicago.

We chose the stories with a fixed process. First we listed some real events from Chicago's recent history that could be seen on the window of time of the dataset (2001-2026) : the COVID lockdown, the Laquan

McDonald scandal, the major blizzards, and the municipal elections. For each one we took the daily crime count in a short window around the event and compared it to a baseline made from the same calendar weeks in two nearby non-event years. We then measured the difference as a percentage. An event became a story only if that percentage was large and clear. Beside just daily crime volume, we also explored other aspect of it like arrest rate or if crime types change differently.

This process worked in two ways. Some events changed the total volume of crime, which is what we expected to find. Others did not change the volume but changed something else, so we had to look closer to see the effect. COVID is the first kind: volume dropped sharply during lockdown. While checking that drop we noticed a spike in late May 2020 that we had not predicted, looked up the date, and found it matched the George Floyd protests. Laquan is the second kind: total crime stayed flat, but the arrest rate fell, so the story is about policing rather than about crime going up or down.

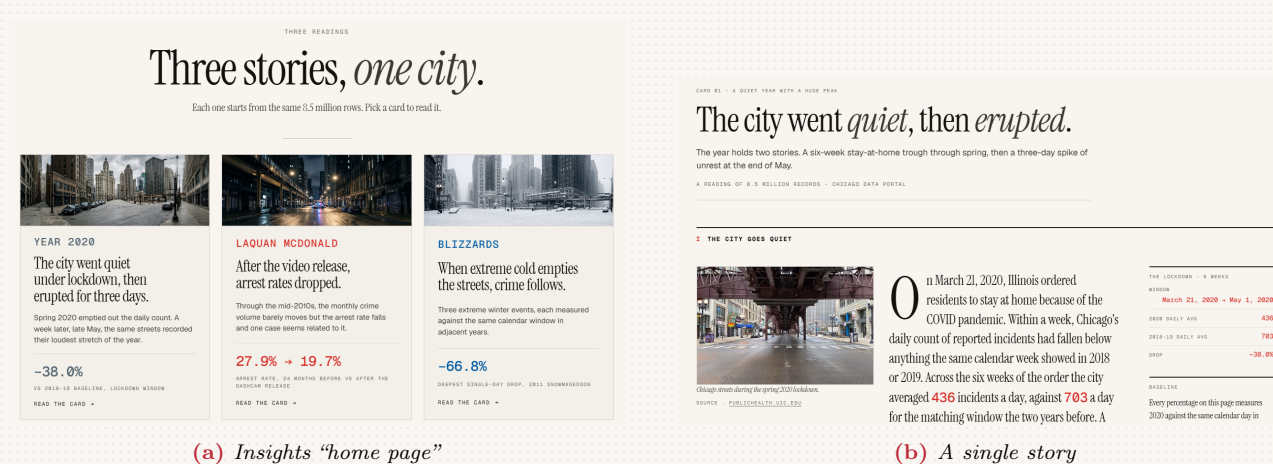


Figure 6: The Insights tab.

Showing the stories. We had to decide how to present each story. The first version put the chart at the top and explained it in text below. This did not read well, because people saw the chart before they knew what to look for. So we reversed the order. The text now leads and asks a question, and the chart answers it. The page uses a magazine layout: text in the main column, figures next to it, and key numbers and short notes in the right margin.

Correlation, not causation. Staying careful with causality was the hardest part to get right. A good story needs a clear point, and the simplest point is always a causal one: the event happened, so the numbers moved. But our data only shows correlation. It records what was reported in a window of time, not why. So on every card we had to write something that read as a story without claiming the event caused the change. The Floyd protests are the clearest case. The data shows some crime categories rising during the late-May 2020 window, and we show that, but our goal is not to argue that the people protesting were violent. The protests were lawful and mostly peaceful, and we say so. We only report what was recorded in those days and leave the cause open. When more than one explanation fit the data, we listed them instead of picking one, and where an effect started before its supposed cause, we pointed that out rather than let the chart imply a link.

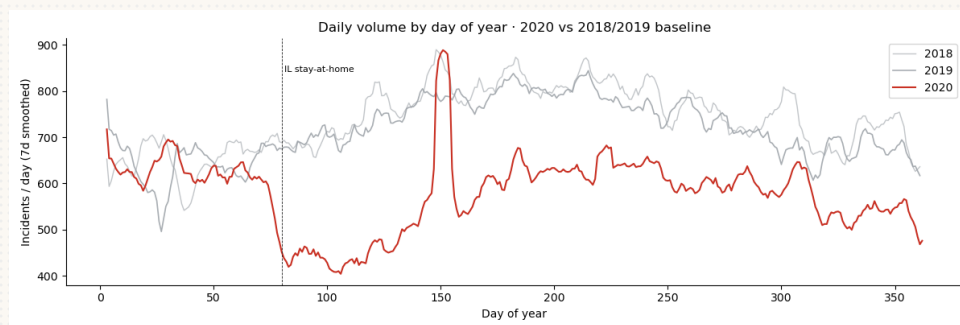


Figure 7: COVID. Volume stays below the 2018/2019 baseline after the Illinois stay-at-home order, with a single spike in late May during the Floyd protests.

The COVID card shows both effects on one chart (Figure 7). Volume fell to about 38% below the 2018/2019 baseline after the Illinois stay-at-home order. Later on the same line there is a sharp spike during the George

Floyd protests, about +272% above the previous two weeks, with a peak near 1,900 incidents in one day. We built it as a main card with a smaller Floyd sub-card.

The Laquan card shows the second type of effect (Figure 8). Total crime volume stayed about the same across the years around the November 2015 video release. The arrest rate, however, dropped from about 28% to about 20%. Because volume held steady while arrests fell, the chart describes a change in policing, not in crime. The timing is also not simple. A federal consent decree followed in January 2019, but the arrest-rate drop began in late 2015, so we present the consent decree and the August 2015 ACLU stop-report agreement as two possible causes and do not choose between them. Figure 8 is the first plot done to explore the evolution of arrest rate around that time to verify the facts of a potential "de-policing" happening after the case, stated by different media articles.

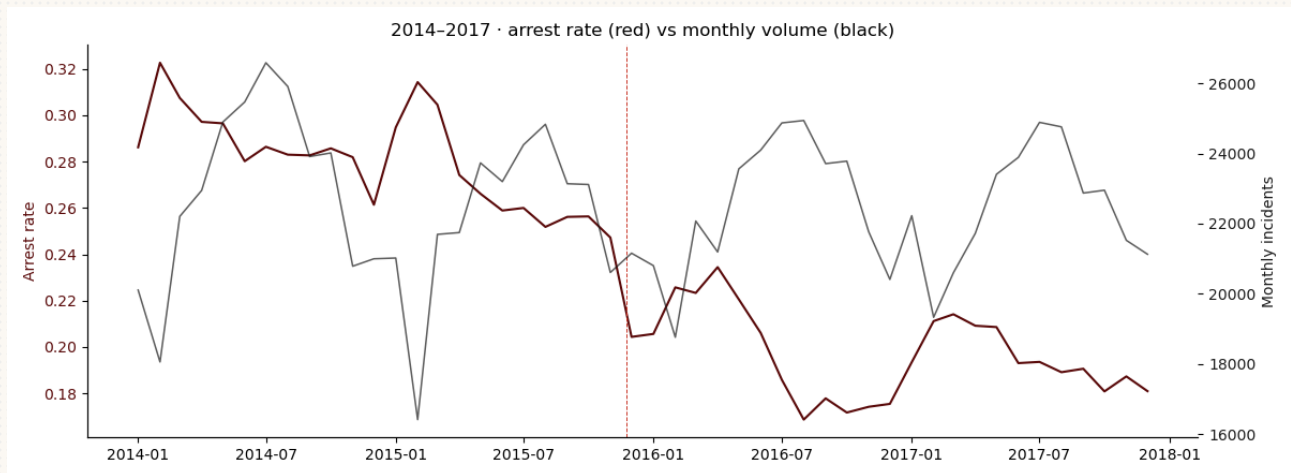


Figure 8: *Laquan McDonald.* Monthly volume (grey) stays steady across the late-2015 video release (dashed line), while the arrest rate (red) drops and stays low.

The blizzards card covers three similar events: the 2011 Snowmageddon storm and the 2014 and 2019 polar vortices. Each one produced a same-day drop in crime, so we built the card around one point: extreme winter weather shuts the city down. The 2011 storm is a useful test because the temperature that week was normal for February, yet the drop was the largest of the three. The cause was the snow, about 21 inches of it, which made the streets hard to use. We also changed one chart during development: the small multiples started as calendar heatmaps of crime, but the colour barely changed from winter to winter, so we switched the calendars to show temperature instead.

Two choices applied to all the cards. The headers use general photos of the city, and we used no images at all around the deaths in the Laquan and Floyd cards. Every large percentage is shown next to its raw count, so a number like "+272%" always appears with the smaller base it came from.

The dropped story: elections. We tested one event that did not become a story. The question was whether reported crime drops in the weeks before a Chicago municipal election when the current mayor is running again. We compared each January–February election window to the same period in non-election years, since all of them fall in low-crime winter.

The result was unclear, which is why we dropped it (Figure 9). Both incumbent years were well below the winter baseline. One open-race control behaved as expected, but the other dropped almost as much as the incumbent years (see Table 1). Three of the four election windows dipped, including one that should not have. With only two cases in each group and a long-term downward trend in the data, the pattern is weak, and reported crime cannot separate a real change from a change in recording. We kept the analysis here as an example of a hypothesis we tested and rejected.

Group	Election (mayor)	Deviation vs. baseline
Test (incumbent)	Feb 2015 (Emanuel)	−35.2%
Test (incumbent)	Feb 2023 (Lightfoot)	−27.7%
Control (open)	Feb 2011	+0.1%
Control (open)	Feb 2019	−28.4%

Table 1: *Pre-vote deviation by election year. Working figures from the notebook*

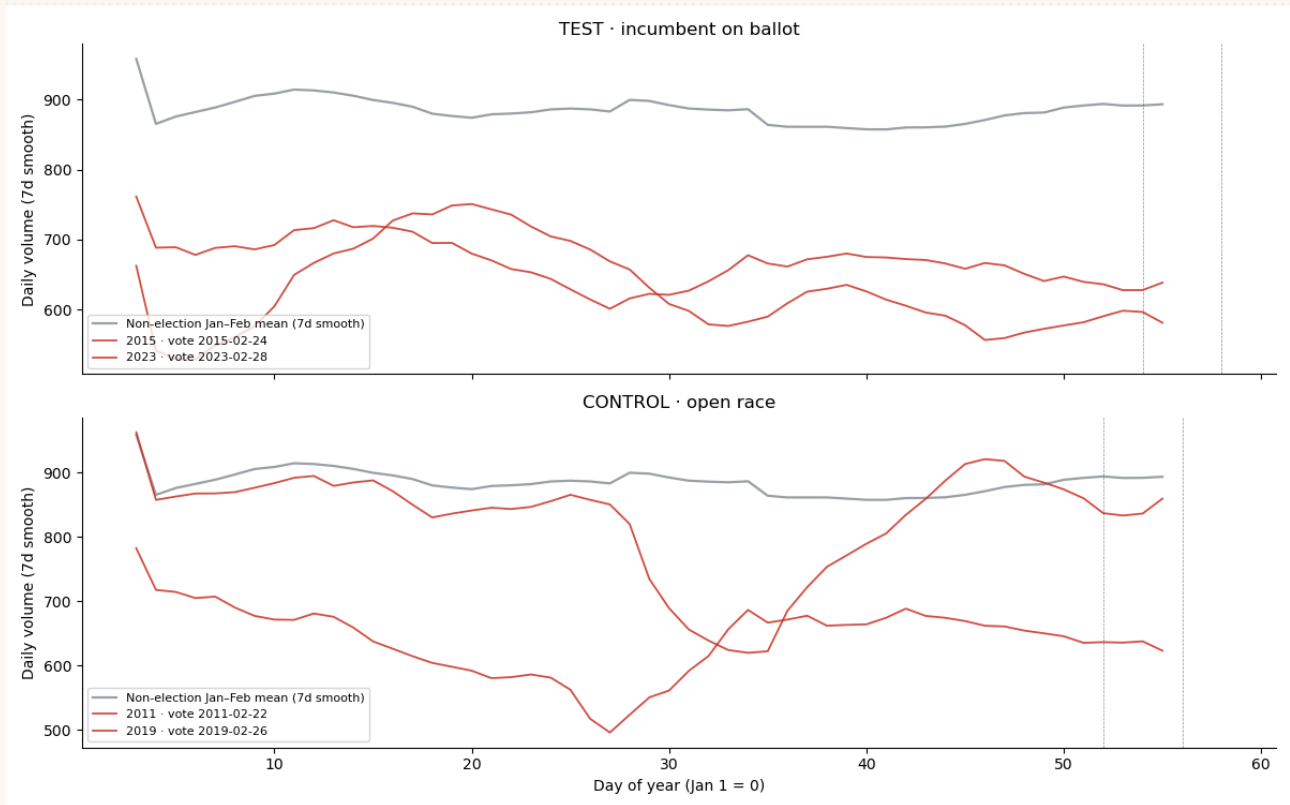


Figure 9: The dropped elections analysis. Top: the two incumbent years (2015, 2023) against the non-election winter mean. Bottom: the open-race controls (2011, 2019).

9 Peer assessment

At the start of the project, most of the work was done by the three of us together. We met regularly for group coding sessions, so that the foundations, the technical implementation and the main design decisions were built and agreed on collectively, by all three of us equally. That shared base also covers the parts the whole site depends on: the exploratory data analysis, the Socrata aggregation pipeline and the per-area data files. Only once it was solid did each of us take the lead on one tab: Max on the Map, Mathis on the Trends, and Julien on the Insights. The process book and the earlier milestones were split evenly across the team, each member writing up the parts they had implemented.

Member	Implementation contributions
Max Henrotin	Map tab + Home page and visual identity + Screencast editing
Mathis Richard	Trends tab
Julien Erbland	Insights tab + Website skeleton and deployment

Table 2: Per-member breakdown of the individual contributions.

10 Final result

The full site is live, and a two-minute screencast walks through the main features:

- **Website:** <https://chicagocrime.vercel.app/>
- **Screencast:** <https://www.youtube.com/watch?v=UxZBNNk30Mc>

The repository, including the data pipeline and this process book, is on GitHub. See the README there for technical usage.